GULI ZHU

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EDUCATION

Beijing Normal-Hong Kong Baptist University (BNBU) 2020.09 – 2024.06
Bachelor of Science in Data Science *Zhuhai, China*

University of Michigan – Ann Arbor 2024.08 – 2026.06
Master of Health Data Science *Ann Arbor, MI*

RESEARCH EXPERIENCE

★ Model Editing for Medical Multimodal LM (BioMed-AI Lab, University of Michigan) 2025.06 – Present
Graduate Researcher (PI: Prof. Liyue Shen) *Ann Arbor, MI*

- Formulated localized factual editing for medical VLMs and designed a **six-dimension** framework centered on **locality vs. generality**, **modality shift**, **compositional edits**, and **temporal stability**.
- Built a fully reproducible evaluation stack (pre/post-edit scoring, automatic grading, evidence attribution, and audit sampling), enabling ablation-driven analysis rather than anecdotal case studies.
- Curated Med-VQA–centric testbeds (VQA-RAD, PathVQA, PMC-VQA, SLAKE) and temporal CT subsets; created stress tests with text/vision perturbations, **cross-modality locality** probes (edit on CT; probe on CXR/MRI/US), and **rare-disease tails**.
- Benchmarked multiple editing approaches across HuatuoGPT-Vision-34B, LLaVA-Med-7B, and related baselines on our 6-dimension benchmark; preparing a public release of the specification and evaluation code. *(Accepted ECCV 2026)*

★ Ultrasound–Diabetes Multimodal Prediction (University of Michigan) 2025.03 – Present
Graduate Researcher (PI: Prof. Liyue Shen) *Ann Arbor, MI*

- Built a unified pipeline linking clinical variables (e.g., BMI, age, labs) with skeletal-muscle ultrasound (deltoid, vastus; SAX/LAX) for diabetes/insulin-resistance prediction.
- Trained clinical-only baselines and developed segmentation (U-Net/MedSAM) to derive quantitative muscle features (echogenicity/density, thickness, cross-sectional area).
- Engineered feature extraction and QC for masks; computed radiomics/texture descriptors; standardized patient-level splits with cross-validated evaluation.
- Designed and compared multimodal fusion strategies (early/late fusion) to predict insulin resistance/T2D; documented metrics and ablation findings.

DIAG Research Subteam – Biomedical Pathology RAG Benchmark (University of Michigan) 2025.01 – Present
Team Lead *Ann Arbor, MI*

- Lead subteam under the Multidisciplinary Design Program (MDP) focusing on retrieval-augmented generation (RAG) for biomedical and pathology LLMs.
- Designed and implemented a pathology-specific RAG benchmark integrating BioLLM and DeepSeek models with modular retrievers and embeddings.
- Developed standardized datasets, retrieval and generation evaluation metrics (e.g., Recall@K, nDCG, faithfulness), and evidence-citation pipelines for benchmarking.

Final Year Project (FYP): Kimera with SLAM semantic mapping and localization (BNBU) 2023.05 – 2023.12
Student Researcher *Zhuhai, China*

- Replicated MIT's Kimera project and explored semantic segmentation in SLAM.
- Investigated impact of semantic segmentation on SLAM localization.
- Introduced optimizations to improve project outcomes.

Large Language Model Evaluation Project (BNBU) 2023.05 – 2023.12
Researcher *Zhuhai, China*

- Conducted review of LLM assessment methods; analyzed pros and cons.
- Evaluated critical thinking of LLMs with psychology/sociology insights.
- Developed methods to assess critical thinking abilities of LLMs.

Genetic Disease Research Project in Data Science (BNBU) 2022.10 – 2023.04
Project Subteam Leader *Zhuhai, China*

- Led data analysis and modeling; trained Decision Trees, Gradient Boosting Trees, Random Forests, and Logistic Regression.
- Applied overfitting techniques to enhance performance after identifying suboptimal recall.
- Logistic Regression achieved accuracy rate of 93%.
- Paper accepted by IEEE MedAI 2023; results implemented in local hospitals.

Research Assistant for Professor Xuemei Chen (BNBU) 2022.01 – 2022.04
Research Assistant *Zhuhai, China*

- Assisted Professor Chen in linguistics research with a focus on data analysis; helped the publication of paper.

Chinese Academy of Sciences ML Online Internship Program 2021.06 – 2021.09
Intern *Remote*

- Constructed a simulation time-series model for Covid-19 using machine learning and data visualization tools.

INTERNSHIP

ByteDance – Machine Learning Intern 2024.03 – 2024.07
Intern, TikTok Endorsement Project *Shanghai, China*

- Developed multimodal recommendation model for TikTok product endorsement.
- Built product recognition (CV) and sentiment analysis (NLP).
- Designed algorithms for clarity analysis, keyframe extraction, clothing item recognition.
- Integrated LLMs to enhance recommendation accuracy and intent analysis.

UIC Mathematics Summer Course Internship (TA) 2021.06 – 2021.08
Teaching Assistant *Zhuhai, China*

- Wrote lesson plans, created courseware, designed course activities.

Guangzhou Institute of Tropical and Marine Meteorology 2022.06 – 2022.08
Research Intern *Guangzhou, China*

- Analyzed typhoon forecasting errors using data-based methods.
- Verified results with multiple models and visualization tools.

PUBLICATION

Zhu, G., Wu, C., & Shen, L. (2026). *Evaluating and Understanding Model Editing for Medical Vision Language Models.* First-author manuscript submitted to ECCV 2026, Accepted.

Wang, D., Huang, S., Cao, J., Feng, Z., Jiang, Q., Zhang, W., Chen, J., Liu, C., **Zhu, G.**, Liao, W., Zhang, L., Guo, W., Liu, L., Yang, J., & Qiuping, L. (2023). *A Multivariate Logistic Regression Model with Over Sampling for Bronchopulmonary Dysplasia Associated with Pulmonary Hypertension in Very Preterm Infants: A Multi-center Retrospective Study.* Proceedings of The First IEEE International Conference on Medical Artificial Intelligence (IEEE MedAI 2023). [Link]

Wang, D., Huang, S., Cao, J., Feng, Z., Jiang, Q., Zhang, W., Chen, J., Liu, C., **Zhu, G.**, Liao, W., Zhang, L., Guo, W., Liu, L., Yang, J., & Qiuping, L. (2024). *A comprehensive study on machine learning models combining with oversampling for bronchopulmonary dysplasia-associated pulmonary hypertension in very preterm infants.* Respiratory Research. [Link]

Vu, K., Maddox, A. M., Tonon, C. W., **Zhu, G.**, Wang, T., Opperman, R. M., Ding, Q., Bai, Z., Liu, Z., Wang, Z., Rao, A., & Yoon, D. (2025, June). *BOARD #92: WIP: Generative AI-based Learning Tutor for Biomedical Data Science (GAIL Tutor BDS).* Proceedings of the 2025 ASEE Annual Conference & Exposition. [Link]

Khan, S., Klochko, C. L., **Zhu, G.**, Ma, D., Wu, C., & Shen, L. (2025). *Early Noninvasive Prediction of Insulin Resistance through Muscle Ultrasound Radiomics.* Radiological Society of North America (RSNA 2025), Scientific Presentations — Oral (accepted), Chicago, IL, Nov–Dec 2025.

OTHER INFORMATION

Institution rename: UIC = BNU–HKBU United International College (now renamed as BNBU).

Honors

- Outstanding Project Award, 2021 ”Jingniu Hui” Cup UIC Innovation and Entrepreneurship Competition
- 2021 UIC Outstanding Volunteer
- Outstanding Volunteer, 7th UIC Fu Chong Youth Entrepreneurship Forum
- Province Second Prize, 2022 National Mathematical Modeling Competition
- Best Poster, UIC Computer Science Exhibition 2023
- Guangdong Medical Valley Scholarship, 2023

Extra-curricular Activities

- Project Manager, Cyber Academy Project team of UIC Enactus (developed 150-hour programming course).
- Undersecretary, Intelligence Capital Department of UIC Enactus.

Computer Skills

Languages: Python, R, SQL, Bash, C
ML/AI: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers,LoRA, PyTorch Lightning, XGBoost
Data & Distributed: pandas, NumPy, Dask, Spark, Hadoop
MLOps & HPC: Docker, Git, Conda, CUDA, Slurm, Weights & Biases, Linux/Unix
Web/Services: FastAPI, Flask, Django, HTML/CSS